

One-Line Challenge

Eckel, TJHSST AI1, Fall 2023

Background & Explanation

This assignment is not about good coding practice, it's about stretching your understanding of the Python programming language to its limits. How much can you accomplish with a single line of code?

Good luck!

Getting Set Up

This assignment is about solving problems from the website Advent of Code in one line each. Advent of Code releases 50 fun programming challenges, 2 per day, during the month of December; I love working on them each year (when I have time!)

This assignment has **extremely precise instructions**, so follow along **carefully**. First we'll get set up, then I'll explain the rules.

- 1) Go to adventofcode.com/2022/ (Make sure you're on the 2022 version.)
- 2) Validate and login in whatever way you choose.
- 3) You'll see an inverted list of the numbers 1 to 25. Each of these is a link to a two-part problem. After solving #1 part A, for example, you'll get access to #1 part B. So there are a total of 50 problems here. You'll need to do both parts of problems 1, 2, 3, and 4, for a total of 8 solutions. (But don't start coding yet.)
- 4) Each problem has input; you access the input by clicking on a link to it in each problem description. Before you begin coding, download the input for each of these four problems **and save each input in a file with these specific names:**
 - "in01.txt"
 - "in02.txt"
 - "in03.txt"
 - "in04.txt"
- 5) Then, make a **single Python file** that will contain all 8 of your solutions. Format this Python file as shown:

```
print("1A:", <your solution to 1A>)
print("1B:", <your solution to 1B>)
print("2A:", <your solution to 2A>)
print("2B:", <your solution to 2B>)
print("3A:", <your solution to 3A>)
print("3B:", <your solution to 3B>)
print("4A:", <your solution to 4A>)
print("4B:", <your solution to 4B>)
```

While you work, I encourage you to make temporary Python files to play with each problem, one by one. Use this file for your final solutions only!

Once you're set up, continue to the rules on the next page.

The Rules

You must solve each of the 8 problems specified on one line each. Let's be very specific.

- Your **full solution** must be **inside the print statement parentheses** on each line. (So, no shenanigans using semicolons to add additional code on the same line after the print statement.)
- You **must read in the input files as named in step 4 above and base your solution on the contents of those files**. (So, no shenanigans where you figure out the solution in a different way and then just hardcode that particular number to print.) In particular, I'll be testing your code on input files with the **same names** but **different contents**, so your code can't assume any specific thing about a particular file, eg # of lines.
- You must have **only the 8 print statements**, no other lines of code before or after. All opening of files, reading of files, and processing of contents must occur in the solutions inside the print statements.
- In particular, this means **you may not import anything at all**, not even sys, since you're hardcoding the file names and no command line arguments are needed.
- This is not a challenge to find creative workarounds to the rules; Python is absurdly adaptable and there are absolutely slick things you can do where you store a bunch of code in a separate text file and read in that text file and execute that code or something like that. That is not the purpose of this challenge; you **may not run any cleverly hidden/disguised/relocated multiline solutions**. This includes something like using "eval()" to run a whole program written with "\n" or ";" in between each line of code in a string. I'm sure there are other ways to do this I haven't thought of; if you're not sure if your solution would break this rule, you can always ask.
- Aside from the banned shenanigans above, there are no other requirements. In particular, your runtime isn't really a concern; if you have an inefficient solution with a lot of redundancy, but it all fits on one line without any shenanigans, it's fine.

Specification

Submit a single Python script to the link given on the course website. When executed, it should run your code for all of the eight problems tasks and output the correct answer for each one based on my modified input files.

This assignment is **complete** if:

- You follow the instructions on the submission form to format your submission properly.
- You follow all the rules (no shenanigans).
- All 8 problems output correct answers based on my modified input files.